PARMCO — AI Interaction Log

October 27, 2026 | MIT App Inventor | Checkpoint 2 preparation

Includes user prompts as written and detailed summaries of the corresponding assistant responses. Assistant responses are paraphrased rather than verbatim; screenshot contents are described, not embedded. No Raspberry Pi testing of the newly added controls occurred during this session.

1. User prompt

Alright lets set up the remainder of the buttons

Assistant response (summary): Outlined remaining controls for the app: Set Speed / RPM, angle target, calibrate zero, maintain mode, and feedback displays. Began with Designer components Label_Speed, TextBox_RPM (NumbersOnly), and Button_SetRPM. Proposed RPM:<value> followed by newline as the Bluetooth message.

2. User prompt

Alright those are in

Assistant response (summary): Explained how to assemble Button_SetRPM.Click: check BluetoothClient1.IsConnected, reject empty TextBox_RPM.Text, and send a join of RPM:, entered text, and a real newline. Suggested testing RPM:100 later; the existing Pi server would return ERR:UNKNOWN_COMMAND until extended.

3. User prompt

The PI isn't with me at the moment. Just continue

Assistant response (summary): Continued offline App Inventor setup. Added Label_Angle, TextBox_Angle, Button_SetAngle, and Button_Zero. Described ANGLE:<value> and ZERO messages with newline terminators and connection checks; clarified that the Pi would not yet move or calibrate the motor.

4. User prompt

K moving on

Assistant response (summary): Explained Maintain Mode toggle: Button_Maintain, global MaintainMode initialized OFF, and MAINTAIN:ON / MAINTAIN:OFF commands. Added proposed Label_CurrentRPM and Label_CurrentAngle readouts for future telemetry.

5. User prompt

K lets continue

Assistant response (summary): Expanded Clock1.Timer receive logic: read a newline-terminated message into global ReceivedMessage, distinguish RPM_ACTUAL: and ANGLE_ACTUAL: prefixes, extract values after the colon, and otherwise display acknowledgments in Label_PiResponse. Suggested using Text starts at compared with 1 and list item 2 after splitting at the colon.

6. User prompt

the starts at block doesn't connect to the if block

Assistant response (summary): Explained that Text starts at returns a numeric position, not a Boolean. Directed user to put it into the left side of a Math equals (=) block and compare with 1 for both RPM_ACTUAL: and ANGLE_ACTUAL:.

7. User prompt

so it should look like this? [screenshot attached]

Assistant response (summary): Reviewed screenshot of Clock1.Timer blocks. Confirmed most structure was correct; suggested adding a degree symbol to the angle label and combining separate checks into an if / else if / else structure so acknowledgments reach Label_PiResponse.

8. User prompt

? [updated screenshot attached]

Assistant response (summary): Reviewed revised screenshot. Identified that set Label_PiResponse.Text was still nested inside the angle branch. Instructed moving it into the final else branch and retaining the angle-degree symbol.

9. User prompt

How do i add the degree symbol?

Assistant response (summary): Explained adding a Text block containing the literal degree symbol (°), then adding a third input to the angle label join block. Example display: Current Angle: 90°.

10. User prompt

got it. what else do i need to add to the app now?

Assistant response (summary): Reviewed app checklist and recommended remaining cleanup: reset RPM/angle/response labels on manual disconnect, improve numeric validation, detect unexpected connection loss, and test later with Pi. Started with Button_Disconnect.Click reset logic.

11. User prompt

All done

Assistant response (summary): Provided RPM and angle validation logic. Suggested rejecting empty/non-numeric RPM and RPM <= 0, and rejecting empty/non-numeric angle. Noted the Pi must independently validate commands and that maximum RPM should follow project requirements.

12. User prompt

Alright next thing

Assistant response (summary): Outlined connection-loss handling using global WasConnected; set true on successful connection and false on manual disconnect. Suggested Clock1.Timer else-if branch to reset status, RPM, angle, response, and Maintain Mode if IsConnected becomes false after previously being true. Warned that IsConnected may not immediately detect remote power loss.

13. User prompt

Alright so the disconnect stuff is checkpoint 3. I just need to get the buttons showing up. From the requirements looking at cp3 for the required buttons, I need: a connect button (already present), start and stop buttons(already present), direction(already present), faster and slower (still needed), desired RPM(already present), angle target(already present), calibrate zero (present not named correctly), and set to zero(not present). I dont need it to be able to control the pi yet, just need it to send a message

Assistant response (summary): Adjusted scope back to preparing visible controls for CP2. Gave instructions for Button_Faster and Button_Slower sending FASTER and SLOWER, renaming Button_Zero to Calibrate Zero and sending CALIBRATE_ZERO, and adding Button_SetZero sending SET_ZERO. These command names were proposed for the later Pi implementation.

14. User prompt

No set_zero is to set the current position of the encoder to zero. This also is just to finish out CP2 on my end

Assistant response (summary): Acknowledged user correction provisionally, reiterated that the immediate goal was button visibility and Bluetooth messages, and asked for clarification about Calibrate Zero versus Set to Zero before finalizing their meanings.

15. User prompt

[Uploaded ECSE4235_PARMCO_Project (2).pdf]

Assistant response (summary): Read the assignment and corrected the interpretation using its explicit definitions: Calibrate Zero stores the current manually aligned blade position as encoder zero; Set to Zero moves the motor back to the stored zero using encoder feedback. Noted calibration is required before set-to-zero or angle-target operation. Also identified logging and over-range controls in the full requirements.

16. User prompt

kk

Assistant response (summary): Provided Designer and Blocks instructions for Button_Logging and Label_Logging, global Logging initialized false, and toggle commands LOG:START / LOG:STOP. Included display updates and reset on disconnect. Noted filename and elapsed-time indicators are required eventually.

17. User prompt

kk

Assistant response (summary): Provided instructions for Button_OverRange and global OverRange, toggling OVERRANGE:ON / OVERRANGE:OFF. Recommended resetting it on STOP and disconnect and applying the assignment normal 10-100 RPM range later. Noted these were planned protocol messages.

18. User prompt

Lets not deal with overrange and RPM validation yet cause thats a CP3 thing i thinl. Is there anything else required for the app by checkpoint 2

Assistant response (summary): Confirmed no more app controls were necessary for CP2 under the uploaded rubric: interface must respond to selections and connect reliably to the Pi. Distinguished separate CP2 tasks: keyboard motor controls, IR RPM sensing, and Test A. Recommended exporting the .aia backup and testing new buttons with Pi when available.

19. User prompt

alright I'll test later with the PI. Can you give me my prompts from today

Assistant response (summary): Compiled the October 27 prompt log in chronological order, including short confirmations, screenshot questions, and the assignment upload. Listed 19 user prompts and summarized the session topics; offered a copyable text log.

20. User prompt

I want your responses as well in the pdf format

Assistant response (summary): Generated this PDF with all 20 prompts from October 27 and detailed summaries of the corresponding assistant responses. This document is a faithful documentation summary, not a verbatim transcript of assistant messages.